

SCI-FLT-FIXED v0.1 Engineering-Conformance Specification

SCI-FLT-FIXED-ENGINEERING-CONFORMANCE v0.1/freeze-candidate

Status: implementation-blind conditional freeze-candidate conformance target; no conformity finding

Scientific owner: Grant Wilson

Stage B date: 2026-08-31

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Document identity: SCI-FLT-FIXED-ENGINEERING-CONFORMANCE v0.1/freeze-candidate
Source SHA-256: 6796518dc6fe3f34558cff01c62bb78deed6e152ee2e01b5f89ae9d0b12c56a9
Shared normative core SHA-256: 7147d242f54d64ca80f6d3a17d309c65f75180457629fed62ce676da93b11089
Author packet manifest SHA-256: 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d
r0.2 owner directive SHA-256: 02ca63dcc8ac19e554fa333c20773ff85264336b4f622024605f6514ec1716d1
r0.3 owner directive SHA-256: c1de5ad5b9f9e5451ec2a3d9d9c18d30f7990eec721497c9d473aaeb2b043551
r0.4 owner directive SHA-256: 855d9709d3fe3b4d71b6a56a020e52ce52164a0384a2ed2d562145f8fb3b0a99
final owner directive SHA-256: c30c49ca19ec15b384ed8178de233e473513fdc520e090c152f571a8a802bf0c
Build recipe SHA-256: 19d4ac5b4238d704a160528f2dc52831ed944cdd48a3f98f1764509ef066ac53
```

Implementation-blind conditional scientific-owner freeze candidate. Owner signature is required. This artifact reports no implementation, validation, calibration, achieved-response, achieved-covariance, numerical-adequacy, performance, readiness, production, or Unity result.

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Normative import: the complete SCI-FLT-FIXED-NORMATIVE-CORE v0.1/freeze-candidate, source SHA-256 7147d242f54d64ca80f6d3a17d309c65f75180457629fed62ce676da93b11089, is incorporated without modification. This view adds no scientific rule. If this view and the imported core differ, the core controls.

1. Conformance target and evidence boundary

This specification translates the shared scientific core into observable engineering obligations without describing, inspecting, or judging an implementation. A future conformance review must bind one candidate realization to one exact core revision, parent generation, resolved operator generation, and product generation.

A claim passes only when positive evidence establishes the exact required identity and behavior. A missing, conflicting, approximate, same-name, or inferred fact yields unavailable or failed status as directed by the core. A finite array, successful process exit, or familiar label is not sufficient.

Conformance, scientific validation, calibration, performance, readiness, and production are separate evidence classes. This specification defines only a future conformance target.

2. Required input and plan records

A candidate realization must accept an immutable request record, effective decision, complete parent binding, and externally resolved fixed plan. The records must expose enough content to reconstruct and compare exact identities, not merely display friendly names.

The parent record must distinguish MAP observation, MAP coadd, and JINC observation roles. It must bind the complete applicable upstream identity, quantity, units, nominal-beam, WCS, support, validity, response, covariance, null, exposure, lifecycle, failure, and provenance state. JINC requires all five atomic numerical roles. Upstream unavailability must remain explicit.

The record must identify exactly one signal vector: the base MAP observation signal, base MAP coadd signal, or normalized `jinc_map`. It must retain exact JINC roles `jinc_signal_numerator`, `jinc_signed_normalization`, `jinc_quadratic_accumulator`, `jinc_map`, and `jinc_coefficient_squared_time`, while proving that only `jinc_map` becomes FLT-SIG. Other map-shaped roles remain separately typed facts or compositions.

The plan record must distinguish requested, effective, disabled, unavailable, and resolved states. A resolved plan must identify FLT-FIXED-CONV as the sole base family, bind one exact sampled convolution, and bind every coefficient, parameter, coordinate domain, grid and metric fact, normalization, support and edge rule, transfer qualification, and provenance fact before application. "Externally resolved" does not transfer FLT policy or application ownership.

The parent record must expose `S_parent_fact`, the typed fact state of every row, and `D_m`, the subset on which an available finite real signal payload exists. It must type $m : D_m \rightarrow R$; parent-shaped storage must not be treated as the scientific vector domain. The output must type $y : S_{out} \rightarrow R$. Membership in `D_m` must be established before a signal payload is evaluated. $k_{\Theta}(r)$, L_{Θ} , and A_{Θ} , J must be real-valued. Each coefficient must be finite, real, unit-typed, canonically exactly represented, and content-bound. Missing, non-finite, complex, unrepresentable, or conflicting coefficients make the plan unavailable before application; a numerical-comparison tolerance may not repair them. A complex transfer $H(\nu)$ remains only a representation of the real operator.

3. Operator reconstruction record

FLT-OPERATOR must provide separate scientific-operator and representation identity subrecords sufficient to reconstruct the exact finite L_{Theta} , frozen J_{full} , and complete A_{Theta} , J . Scientific identity includes:

- ordered D_m and S_{out} scientific domains and their relation to $S_{\text{parent_fact}}$;
- exact WCS, frame, topology, metric, shape, indexing, and pixel-area facts;
- canonical offset-to-coefficient relation, exact values and units, and scientific-operator digest;
- representation-invariant $K_{\text{geom_science}}$, K_{nonzero} , ordinary-method $K_{\text{req}} = K_{\text{nonzero}}$, canonical exact-zero representation, center, extent, tie, phase, subpixel convention, orientation, and handedness;
- declared normalization, DC gain, and distinct signed, absolute, squared, and geometric support summaries;
- full-footprint predicates, exact independent $S_{\text{out_zero}}$, and cause vocabulary;
- qualified transfer facts or explicit unavailability; and
- immutable plan and scientific-operator generation identifiers.

The representation subrecord must separately bind nonauthoritative K_{store} , dense/sparse/cropped/padded encoding, field and byte ordering, compression or container, representation digest, and representation generation.

The zero-operator record must bind $K_{\text{nonzero_zero}} = \text{empty set}$, $K_{\text{req_zero}} = \text{empty set}$, and independently constructed $S_{\text{out_zero}}$ equal to the exact admitted finite parent-signal row domain under its request and predicates. It must not describe a parent row domain as a K_{req} domain or admit arbitrary storage rows by vacuity. A nonzero convolution is exactly one whose K_{nonzero} is not empty.

The record must show that one product applies this exact sampled convolution once and makes no intermediate or reordered-composition claim. An alternate computational mechanism is comparable only after reconstructing the identical declared scientific operator. A representation-only change may create a new representation artifact or generation but must preserve scientific operator identity, FLT product identity, scientific generation, and behavior. A change to a canonical coefficient or other scientific fact is not representation equivalence and creates a new scientific transformation.

4. Admission and application gate

Bundle admission must evaluate `SCI-FLT-FIXED:input_bundle_admission@1` against one exact request, parent, and resolved plan. Parent-row admission must evaluate

`SCI-FLT-FIXED:input_parent_row_admission@1` for every exact parent row and named FLT use. Each profile returns the typed request, applicability, eligibility, missing/conflict, action, and cause state defined by its bound profile. Neither performs payload arithmetic or authors output-row support.

Application may proceed only for an eligible resolved route. Plan state must be independent of parent amplitudes. The selector must be resolved once before convolution arithmetic from declared immutable $S_{\text{parent_fact}}$, row membership, exact input admission, typed facts defining D_m , support, and required predicates. Establishing an authorized required location's membership in D_m is structural screening and must not evaluate m_q or tune the plan. For every output row, FLT verifies all K_{req} locations against those exact facts. The scientific row domain must equal the passing set. Response perturbations, covariance draws, noise realizations, and NOI members must reuse that selector; a member failure produces row unavailability, not re-selection.

No extension, periodic wrap, truncation, local renormalization, inpainting, reflection, clamp, mirror, padding-based admission, edge completion, or value replacement is conforming in v0.1. Rows failing the full-footprint gate remain unavailable with causes, even if a storage payload is finite.

Dense, sparse, cropped, and zero-padded encodings of one canonical kernel may have different representation identities but must reconstruct identical scientific operator and FLT product identities, scientific generation, support, row admission, response, and covariance. Only representation facts, including `K_store`, may differ. Ordinary convolution arithmetic must traverse exactly $K_{\text{nonzero}} = K_{\text{req}}$, not $K_{\text{geom_science}}$. An exact-zero offset creates no arithmetic, payload, influence, covariance, or admission dependency, and its parent payload must not be evaluated or dereferenced.

A requested nonzero convolution with empty `S_out` must retain its application evidence as `applied_no_scientific_output_support` and construct a complete `no_output_support_candidate` with the required attempted-domain proof and no realized FLT-SIG or atomic bundle. For publication, request must be requested, applicability applicable, eligibility ineligible, realization not_produced, and cause `no_full_footprint_output_rows`. It must not relabel the state as not requested, disabled, failed execution, decision unavailable, identity, or zero.

5. Signal, response, transfer, mode, and influence records

Application must compute FLT-SIG from the exact resolved operator on the exact scientific row ordering. The output-unit record must derive units from parent and coefficient units and retain the originating nominal-beam and calibration lineage without introducing a filtered-beam label.

Each response record must name fixed-state linear, already realized parent- grid, parent full-procedure finite difference with FLT fixed, or FLT re- resolved procedure family. The first three must apply the identical frozen `A_Theta, J` exactly once; the fourth is outside v0.1. A full-procedure difference must bind compatible baseline and perturbed membership, availability, WCS, quantity, and support on the frozen required domain. A mismatch retains the parent state change, does not re-resolve `J_full`, and makes affected response rows unavailable. Missing basis, domain, or parent-response identity is explicitly unavailable. The exact zero operator records its local zero derivative without promoting it to an unavailable complete source response. Kernel-only, approximate, multiply applied, differently centered, differently normalized, or differently edged surrogates do not pass.

Transfer and mode records must bind the exact coordinate domain, WCS metric, transform sign and normalization, coordinate/frequency units, origin, ordering, signed/Nyquist treatment, frequency grid, response quantity, attenuation units, band geometry, phase branch, finite-grid domain, null, invariant, anisotropy, and attenuation facts that are scientifically defined. They must distinguish sampled-kernel transfer from the complete row-restricted operator. Unsupported facts remain unavailable. Influence exposes the parent-row coefficient relation; FLT-EXPOSURE-LINEAGE binds parent exposure identity or typed absence and confirms that neither influence nor a convolved exposure plane is physical exposure.

6. Covariance and NOI records

For an available compatible parent covariance, a covariance-qualified candidate must realize the exact two-sided frozen-operator propagation on `S_out`. Its record must state parent stochastic authority separately from output representation and name the conditional model, omitted terms, excluded selection/kernel/beam/WCS/calibration/model uncertainties, domain, ordering, rank, null space, and supported operations.

If an explicitly diagonal parent covariance is propagated, any complete output representation must include induced cross-row covariance. A marginal-only representation must say so. Missing cross terms may not be encoded as zero or independence. For marginal-only parent authority, a row with exactly one nonzero coefficient has the exact conditional marginal $A_{ij}^2 \text{Var}(m_j)$; a row mixing two or more parent variables remains unavailable or explicitly partial. The one-sparse result authorizes no cross-row covariance or independence. The exact-zero row keeps its separately typed zero parent-payload contribution. Structured or partial authority permits only proved-exact operations. Any diagonal-contribution diagnostic must avoid variance, covariance, uncertainty, and precision labels.

FLT-NOI-COMPATIBILITY must be immutable publication-time FLT state containing no future NOI identity. A later NOI child references FLT; an optional reverse relation is separately versioned and cannot mutate FLT.

Every admitted NOI member must receive the identical frozen A_Theta, J; member footprint failure is unavailable. The route rejects filtering uncertainty-like products, approximate transfer, relocation, commutation, substitution, and any per-member re-resolution.

A not_requested_at_FLT_publication state is historical provenance only. It must not block a later independently requested compatible NOI child, whose request, applicability, eligibility, realization, generation, and failure are child-owned and cannot mutate FLT.

7. Atomic publication and lifecycle records

Publication policy must evaluate SCI-FLT-FIXED:output_publication@1 against one exact complete_publication_disposition_candidate. A product_candidate contains all required roles from the imported core, including request-specific honest unavailable companion records. A no_output_support_candidate contains the exact request, parent, plan, operator, K_req, attempted output- domain construction, empty-domain proof, row/cause accounting, application generation, applied-no-support state, and prescribed cause, but no realized FLT-SIG and no atomic bundle. A partial candidate, placeholder, or inferred companion fails publication. The policy defines disposition and prescribed action. VAL may produce a decision artifact but performs no publication. The FLT publisher performs or declines the action and owns final realization and local validity.

Lifecycle evidence must distinguish not_requested, requested, effective, disabled, unavailable, resolved, applied, applied_no_scientific_output_support, complete_publication_disposition_candidate, product_candidate, no_output_support_candidate, publication_decision, not_produced, realization_failed, failed, realized_identity, realized_zero, realized, and superseded. Disabled is not produced. Identity and zero use the same candidate and decision sequence. Required failure propagates and emits no complete product.

Every product must bind immutable parent, plan, scientific operator, representation, output, companion, cause, lifecycle, failure, and provenance generations. A scientific-identity change creates a new scientific generation; a representation-only change does not. A later NOI child or reverse relation is separate and cannot mutate an existing FLT bundle.

8. Requirement conformance matrix

The following matrix routes every stable core requirement to a future observable conformance decision. The exact normative text remains in the imported core.

- SCI-FLT-FIXED-REQ-001: verify package, tranche, convolution-only base, and qualified low-pass subtype identity; reject arbitrary-linear or inference- bearing identity.
- SCI-FLT-FIXED-REQ-002: reconstruct A, S_parent_fact, D_m, and S_out; verify typed partial functions and no additive term or dependency.
- SCI-FLT-FIXED-REQ-003: verify exactly one supported immutable parent role.
- SCI-FLT-FIXED-REQ-004: verify complete applicable parent fields and atomic JINC roles.
- SCI-FLT-FIXED-REQ-005: verify upstream unavailability remains fail-closed.
- SCI-FLT-FIXED-REQ-006: verify observation, coadd, and JINC successor identities and absence of FLT coaddition or inferred commutation.
- SCI-FLT-FIXED-REQ-007: verify all plan and selector state predates payload arithmetic, plan state is amplitude-independent, selector structural screening uses only authorized parent facts, D_m membership precedes payload evaluation, and companions do not re-resolve the selector.
- SCI-FLT-FIXED-REQ-008: compare exact WCS and grid facts and reject approximate joins or resampling.
- SCI-FLT-FIXED-REQ-009: reconstruct separate scientific-operator and representation identities, digests, generations, support sets, and domains; compare exact representation invariance.

- SCI-FLT-FIXED-REQ-010: verify one-time convolution construction from the exact sampled kernel over $K_{\text{nonzero}} = K_{\text{req}}$, without geometric exact-zero, intermediate, or reordered terms.
- SCI-FLT-FIXED-REQ-011: verify every low-pass transfer fact or mark only the qualification unavailable.
- SCI-FLT-FIXED-REQ-012: recompute the full-footprint row set from $S_{\text{parent_fact}}$, D_m , $K_{\text{req}} = K_{\text{nonzero}}$, and predicates independently of geometric or storage zeros, including membership-before-evaluation.
- SCI-FLT-FIXED-REQ-013: verify all excluded rows are unavailable with causes, never promoted by storage shape.
- SCI-FLT-FIXED-REQ-014: detect and reject every deferred edge or replacement method.
- SCI-FLT-FIXED-REQ-015: verify transformed-amplitude identity and exact unit derivation without amplitude-category relabeling.
- SCI-FLT-FIXED-REQ-016: verify nominal-beam and CAL lineage retention and absence of unsupported beam or flux claims.
- SCI-FLT-FIXED-REQ-017: identify the response family, compare its one-time frozen-operator composition, or verify honest unavailability.
- SCI-FLT-FIXED-REQ-018: verify finite-domain transfer, mode, and phase facts or explicit unavailable states without whole-chain promotion.
- SCI-FLT-FIXED-REQ-019: compare influence to coefficients, verify it is not exposure, and verify exact exposure-lineage state.
- SCI-FLT-FIXED-REQ-020: verify distinct computability, support, admission, validity, confidence, and downstream states.
- SCI-FLT-FIXED-REQ-021: compare covariance to exact two-sided propagation.
- SCI-FLT-FIXED-REQ-022: verify separate parent-authority and output- representation axes, conditional model, exclusions, and unknown cross-term handling.
- SCI-FLT-FIXED-REQ-023: test induced off-diagonal terms and reject marginal planes presented as complete covariance.
- SCI-FLT-FIXED-REQ-024: verify empirical uncertainty and significance are absent from FLT ownership.
- SCI-FLT-FIXED-REQ-025: compare every NOI member's full operator state to the signal state.
- SCI-FLT-FIXED-REQ-026: detect and reject per-member selection or re-resolution.
- SCI-FLT-FIXED-REQ-027: verify every atomic role, exposure lineage, allowed honest unavailable record, and immutable NOI-compatibility role without a future NOI identity or completion dependency.
- SCI-FLT-FIXED-REQ-028: verify both complete publication-disposition candidate variants before decision, including causes, failures, and generation bindings.
- SCI-FLT-FIXED-REQ-029: test distinct disabled, identity, and zero outcomes.
- SCI-FLT-FIXED-REQ-030: mutate scientific and representation facts separately; verify only scientific changes create a new FLT identity and scientific generation, and verify NOI nonmutation.
- SCI-FLT-FIXED-REQ-031: remove or conflict each required fact and verify fail-closed behavior without fallback.
- SCI-FLT-FIXED-REQ-032: evaluate distinct bundle and exact parent-row admission request, applicability, eligibility, unavailable, exclusion, and cause branches while FLT constructs output support.
- SCI-FLT-FIXED-REQ-033: evaluate both exact publication-candidate variants, disposition, axes, prescribed action, honest-unavailable, partial, disabled, identity, zero, and failed branches; verify the publisher acts.

- SCI-FLT-FIXED-REQ-034: verify VAL only binds or evaluates an immutable approved policy and authors no FLT fact, arithmetic, publication, or realization.
- SCI-FLT-FIXED-REQ-035: verify no generic downstream admission follows from product availability.
- SCI-FLT-FIXED-REQ-036: verify every excluded family and every Stage B nonclaim remains outside the product claim set.
- SCI-FLT-FIXED-REQ-037: freeze J_full once and verify identical selector reuse, plan amplitude-independence, S_parent_fact/D_m screening before payload evaluation, and exclusion of unsupplied selection uncertainty.
- SCI-FLT-FIXED-REQ-038: verify K_geom_science, K_store, K_nonzero, ordinary K_req = K_nonzero, exact-zero, identity, and zero's empty support sets with independently constructed S_out_zero.
- SCI-FLT-FIXED-REQ-039: evaluate all three typed policy objects and each request-specific companion qualification.
- SCI-FLT-FIXED-REQ-040: distinguish every response family and prevent local zero derivative from becoming an unsupported complete response claim.
- SCI-FLT-FIXED-REQ-041: verify exactly one sampled convolution application and absence of composition, collapse, or reordering claims.
- SCI-FLT-FIXED-REQ-042: verify preregistered numerical comparison fields, frozen bounds, independent oracle, and nonclaim boundary.
- SCI-FLT-FIXED-REQ-043: verify exact inherited exposure lineage and reject influence or convolved exposure as physical exposure.
- SCI-FLT-FIXED-REQ-044: verify external timing while retaining exact FLT-owned plan, selector, application, and publication ownership.
- SCI-FLT-FIXED-REQ-045: verify only the exact parent signal role becomes FLT-SIG and all JINC diagnostics, support, validity, exposure, response, and covariance retain their typed routes.
- SCI-FLT-FIXED-REQ-046: exercise every covariance-authority row and reject unproved marginal, structured, partial, or unavailable strengthening while admitting only the exact one-sparse conditional marginal edge case.
- SCI-FLT-FIXED-REQ-047: inspect immutable FLT-NOI-COMPATIBILITY content and prove a publication-time not-requested state does not block a later independent child and that children and reverse relations do not mutate FLT.
- SCI-FLT-FIXED-REQ-048: verify bundle/parent-row profile decisions, FLT selector construction, both publication candidate variants and axes, VAL decision-artifact boundary, and publisher action.
- SCI-FLT-FIXED-REQ-049: compare every bound low-pass transform convention and reject a complete but different convention.
- SCI-FLT-FIXED-REQ-050: compare full-procedure parent domains and verify mismatch unavailability without selector re-resolution.
- SCI-FLT-FIXED-REQ-051: reproduce every authority-manifest path, byte count, SHA-256, role, authority state, compatibility or supersession state, and generated-view relation or route the dependent claim to unavailability.
- SCI-FLT-FIXED-REQ-052: verify $m : D_m \rightarrow R$, $y : S_{out} \rightarrow R$, and reject every missing, non-finite, complex, unrepresentable, or conflicting coefficient before application without tolerance repair.
- SCI-FLT-FIXED-REQ-053: exercise empty S_out for a requested nonzero convolution and verify the exact no-output-support candidate, four axes, cause, retained evidence, and forbidden relabelings.

9. Falsifiable prediction suite

Each future prediction evaluation must bind exact inputs, operator state, expected row domain, expected companions, expected lifecycle, comparison semantics, and causes before execution. The normative expected outcomes are in the imported core.

- SCI-FLT-FIXED-PRED-001: identity operator and identity lifecycle.
- SCI-FLT-FIXED-PRED-002: zero operator with empty arithmetic/dependency sets, independent S_{out_zero} , local zero derivative and covariance, and honest unavailable complete response.
- SCI-FLT-FIXED-PRED-003: scalar input linearity on one frozen selector.
- SCI-FLT-FIXED-PRED-004: constant input under exact DC gain and every authorized normalization.
- SCI-FLT-FIXED-PRED-005: impulse, center, orientation, phase, indexing, and exact-nonzero arithmetic support.
- SCI-FLT-FIXED-PRED-006: exact separately typed response-family composition with one frozen operator application.
- SCI-FLT-FIXED-PRED-007: signed coefficients and distinct support summaries.
- SCI-FLT-FIXED-PRED-008: zero-sum constant-mode null without claim promotion.
- SCI-FLT-FIXED-PRED-009: typed-domain full-footprint inclusion, dependent- row removal, representation invariance, and exact-zero no-dereference.
- SCI-FLT-FIXED-PRED-010: rejection of every deferred edge and replacement method, including support-conditioned renormalization.
- SCI-FLT-FIXED-PRED-011: S_{parent_fact}/D_m exclusion of missing, unavailable, non-admitted, and non-finite required locations.
- SCI-FLT-FIXED-PRED-012: exact complete covariance on ordered S_{out} .
- SCI-FLT-FIXED-PRED-013: induced cross-row covariance conditional on an explicit independent-diagonal parent model.
- SCI-FLT-FIXED-PRED-014: unavailable parent response or covariance and rejection of surrogates.
- SCI-FLT-FIXED-PRED-015: exact WCS and grid mismatch failure.
- SCI-FLT-FIXED-PRED-016: distinct observation, coadd, and JINC identities with no FLT coadd product.
- SCI-FLT-FIXED-PRED-017: exact frozen-selector NOI member parity and member-level unavailable footprint state.
- SCI-FLT-FIXED-PRED-018: per-member re-resolution rejection.
- SCI-FLT-FIXED-PRED-019: disabled, identity, zero, and failure lifecycle distinctions.
- SCI-FLT-FIXED-PRED-020: upstream unavailable MAP and JINC route.
- SCI-FLT-FIXED-PRED-021: low-pass qualification completeness.
- SCI-FLT-FIXED-PRED-022: exact-zero support plus distinct dense/sparse representation identities with identical scientific and product identity.
- SCI-FLT-FIXED-PRED-023: zero-operator empty support sets and independent admitted finite parent-signal output domain.
- SCI-FLT-FIXED-PRED-024: independent exact sampled low-pass transfer under the identical complete transform convention.
- SCI-FLT-FIXED-PRED-025: rejection of every non-signal parent role as ordinary-method FLT-SIG.

- SCI-FLT-FIXED-PRED-026: filtered-variance sensitivity to parent cross terms with identical marginal diagonals plus one-sparse, two-sparse, and exact-zero marginal-only fixtures.
- SCI-FLT-FIXED-PRED-027: later NOI child and reverse-relation nonmutation of the immutable FLT bundle, including a late request after publication-time not-requested provenance.
- SCI-FLT-FIXED-PRED-028: full-procedure response domain incompatibility and frozen-selector preservation.
- SCI-FLT-FIXED-PRED-029: complete no-output-support candidate, retained evidence, exact publication axes and cause, and forbidden relabelings.
- SCI-FLT-FIXED-PRED-030: invalid-coefficient rejection before application with no numerical-tolerance repair.

10. Numerical-comparison evidence record

A future candidate comparison must bind one immutable policy before results are observed. The evidence record must identify its independent exact or high- precision oracle; absolute, relative, and near-zero rules; signed-cancellation and zero-sum cases; conditioning and operation-count dependence; covariance comparison; sequential and parallel agreement; overflow, underflow, and non- finite handling; simultaneous row decisions; lifecycle; and provenance.

The record must prove that bounds were not changed after a failure and that scientific coefficients and identity remained exact. Passing such a policy would be future conformance evidence only. This specification supplies no candidate result, numerical-adequacy finding, validation, or performance claim.

11. Traceability, identity, and reproducibility record

TRACEABILITY.json maps every requirement and prediction identifier to its normative-core section, scientist-rationale section, engineering-conformance section, exact admitted Stage A object sections, and exact r0.2, r0.3, r0.4, and final owner-directive sections where a revision changes semantics. The durable verifier must reject missing, duplicate, extra, malformed, or unadmitted trace entries.

BUILD_BINDING.json records the packet manifest binding, all 17 admitted object hashes and byte counts, exact r0.2, r0.3, r0.4, and final owner- directive bindings, every Stage B source and build-tool hash, deterministic build environment, output PDF identities, page counts, byte sizes, and SHA-256 digests. The source-closure, owner-decision parity, core/view parity, semantic-change, profile, rebuild, and visual-QA records are required. A clean separate rebuild must reproduce the three PDF digests exactly.

AUTHORITY_MANIFEST.json is the single conditional-freeze-candidate entry point. It binds the Stage A packet, all four directives, boundaries, policy and numerical drafts, traceability, build binding, closure/parity/change/profile/rebuild/ visual reports, all sources, tools, and PDFs by exact path, bytes, SHA-256, scientific/process/representation role, authority state, compatibility or supersession state, and generated-view relation. Its own external digest provides the single entry point without a self-hash.

PDF text must expose the document identity, exact source digest, exact shared core digest, exact packet-manifest digest, and exact builder digest. PDF page rendering and visual QA are required in addition to content and hash checks.

12. Nonclaims

This conformance specification does not identify, inspect, modify, or judge an implementation. It records no implementation conformity, algorithm change, validation result, calibration state, achieved response or covariance, numerical adequacy, performance, readiness, scientific freeze, production status, or Unity activity.